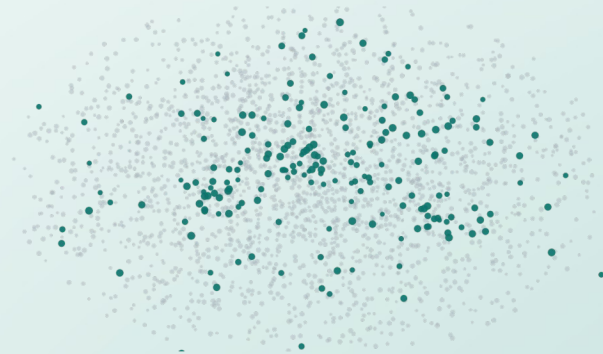


UNLV 2026 SUMMER · PROJECT 5

Comparison of LASSO-based Feature Selection for High-Dimensional Genomic Data

From LASSO to Hi-LASSO — which one picks the right genes?



each dot = a gene · only a few are disease-linked

Thousands of genes, only tens of patients



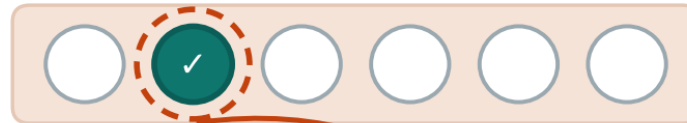
Normal regression overfits here — **and only a few genes truly matter.**

Correlated genes, and an unstable pick

Correlated genes → LASSO's pick keeps changing

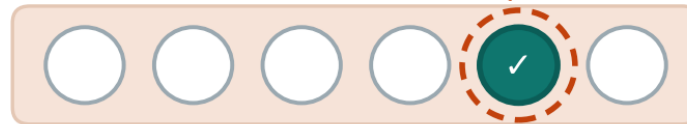
6 highly-correlated genes (rise & fall together)

Run 1



pick jumps

Run 2



Same correlated genes, tiny data change → a different gene is selected (unstable)

LASSO keeps one gene from a correlated group — and the choice flips run to run.

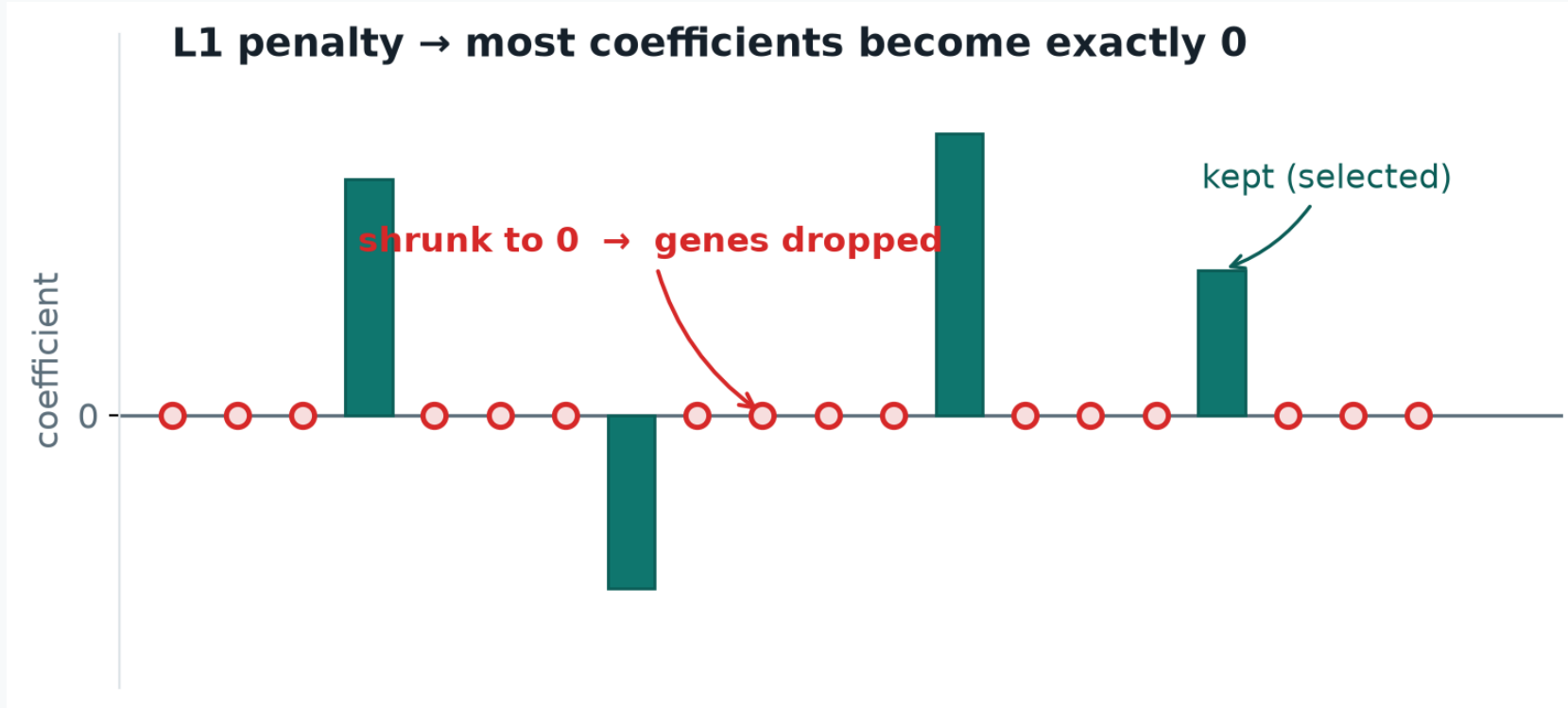
Four methods, each fixing the one before

Each arrow = the problem the next method fixes



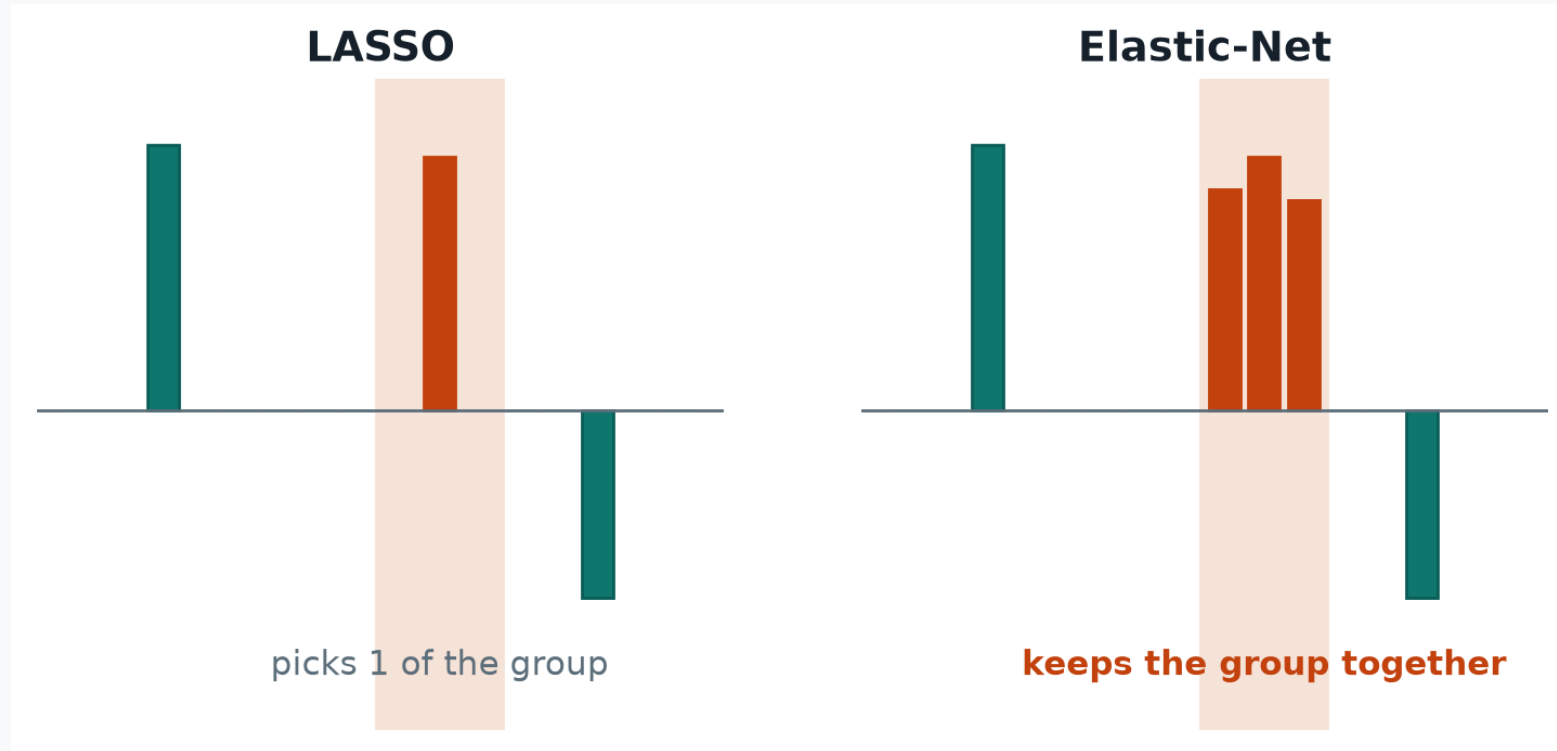
Each arrow is the problem the next method solves. **We built and compared all four.**

L1 penalty zeroes out most genes



A nonzero coefficient = a selected gene. **Simple, but weak on correlated genes.**

Keeps correlated genes as a group



L1 + L2 selects a correlated group together — **but it is still one single fit.**

Vote over many random subsets

Each random subset selects a few genes — some keep coming back

	g1	g2	g3	g4	g5	g6
Subset 1	☒	☒	☐	☒	☐	☐
Subset 2	☐	☒	☐	☒	☒	☐
Subset 3	☐	☒	☒	☒	☐	☐
Subset 4	☒	☒	☐	☒	☐	☒
Subset 5	☐	☒	☐	☒	☐	☐
times picked	2	5	1	5	1	1
Genes picked in many subsets → SELECTED (stable, can exceed n)						

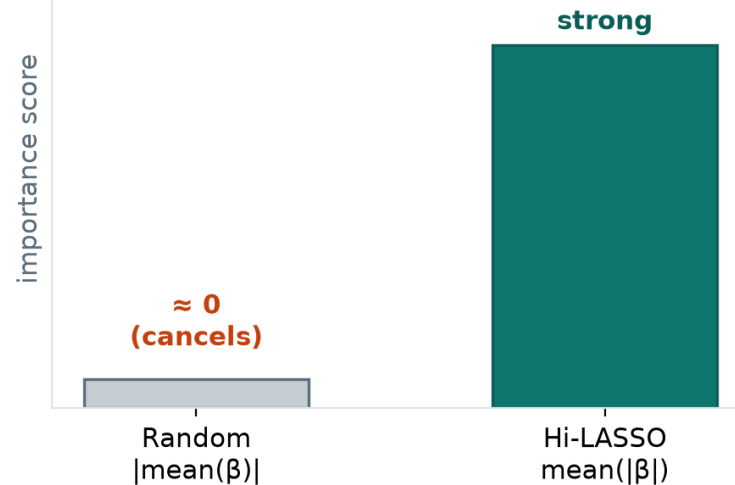
Averaging over subsets makes selection stable — and can pick more genes than samples.

Two math fixes make Hi-LASSO reliable

Hi-LASSO's two math fixes: average $|\beta|$, and count only when selected

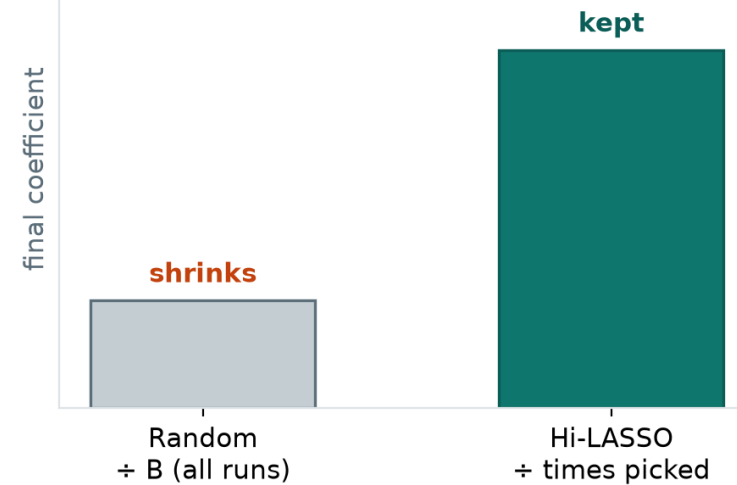
Fix 1 · opposite signs cancel

β flips $+/-$ across bootstraps



Fix 2 · rare picks diluted

selected in 3 of 10 runs



Average $|\beta|$ so signs reinforce, and count only when a gene is picked — **the most reliable selection in our tests.**

Four datasets with a known answer key

Four synthetic datasets — we know the true genes (answer key)

Dataset I

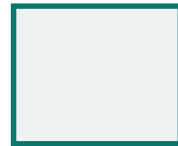
$p=100 \cdot n=50$



10 true genes

Dataset II

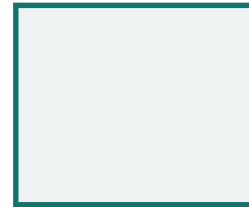
$p=1,000 \cdot n=100$



50 true genes

Dataset III

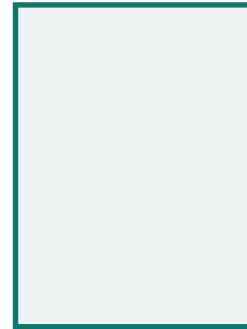
$p=10,000 \cdot n=200$



50 true genes

Dataset IV

$p=10,000 \cdot n=400$



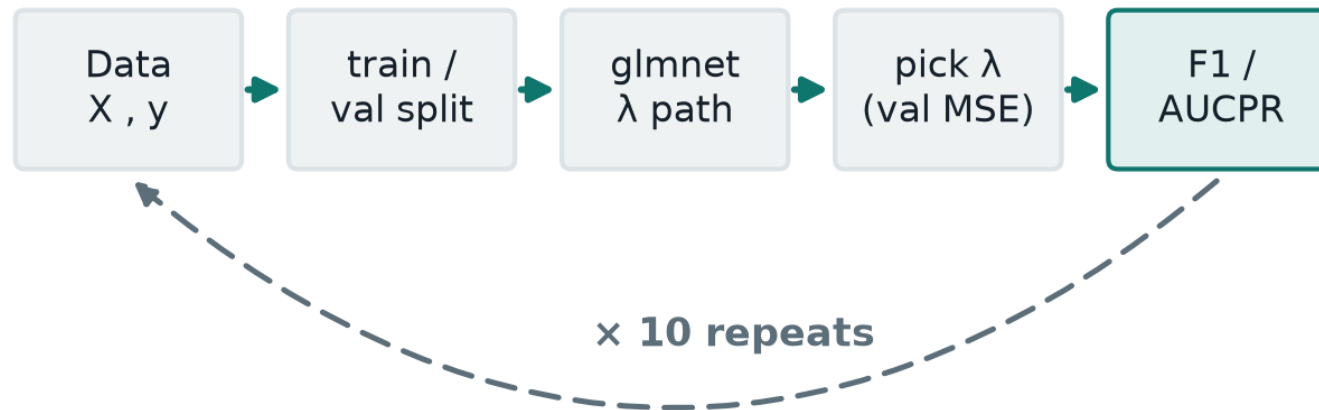
50 true genes

$y = X\beta + \text{noise}$ · β fixed (seed 34), only X & noise change each run

We generate the data, so we know the true genes — **that lets us score with F1.**

One fair protocol, ten repeats

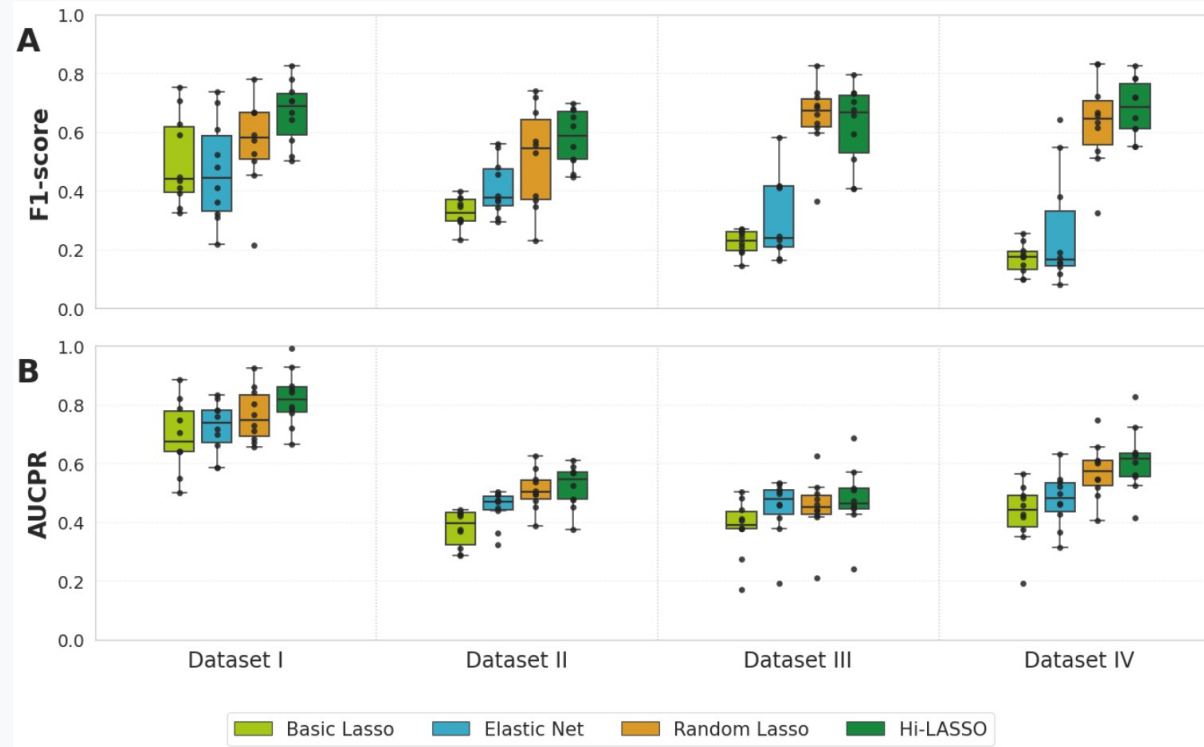
Same protocol for every method — fair comparison



Random / Hi-LASSO bootstrap: $q = n$, $B = (p / q) \times 30$

Same steps for every method: **pick λ on validation, repeat 10 times.**

Feature selection performance



Random Lasso and Hi-LASSO clearly lead — and the gap grows on the hardest datasets (III–IV).

How the four methods compare (F1 & AUCPR)

Bootstrap methods (bottom two rows) stay strong; Hi-LASSO best overall

F1 — did it SELECT the right genes?

Basic Lasso	0.50	0.33	0.23	0.17
Elastic Net	0.47	0.41	0.30	0.26
Random Lasso	0.56	0.51	0.65	0.63
Hi-LASSO	0.66	0.58	0.62	0.68
	Data I (n50,p100)	Data II (n100,p1k)	Data III (n200,p10k)	Data IV (n400,p10k)

AUCPR — did it RANK the right genes?

Basic Lasso	0.69	0.38	0.38	0.43
Elastic Net	0.72	0.45	0.45	0.48
Random Lasso	0.76	0.51	0.45	0.57
Hi-LASSO	0.82	0.52	0.48	0.61
	Data I (n50,p100)	Data II (n100,p1k)	Data III (n200,p10k)	Data IV (n400,p10k)

Bootstrap methods (bottom two rows) stay dark on both metrics — **Hi-LASSO is best overall.**

Top genes on real data — 3 are already known

Applied Hi-LASSO to real GBM / LGG data:
266 patients · 19,777 genes · survival time.

No ground-truth β , so we rank the top 10 genes by $|\text{coefficient}|$.

3 of the 10 (✓) are already known glioma genes — the other 7 are new candidate biomarkers.

Gene	Coefficient	Correlation	Glioma link
DEFB120	0.1152	0.5019	—
ONECUT3	0.1044	0.4761	✓ ref [5]
GSC2	0.0914	0.4510	—
COL22A1	-0.0820	-0.3025	✓ ref [9–11]
UGT1A10	0.0708	0.3798	—
AMTN	0.0643	0.4520	—
LINC00442	0.0603	0.4495	—
HILS1	-0.0600	-0.2752	✓ ref [6–8]
USP17L6P	0.0570	0.3759	—
GPR50	0.0564	0.3607	—

Three things that were hard



Results jumped every run

The same experiment gave different numbers each time.

→ Froze the random seed — only X & noise change.



Bootstrap was expensive

$p = 10,000$ means thousands of LASSO fits per run.

→ Long runtimes — ran the 10 repeats in batches.



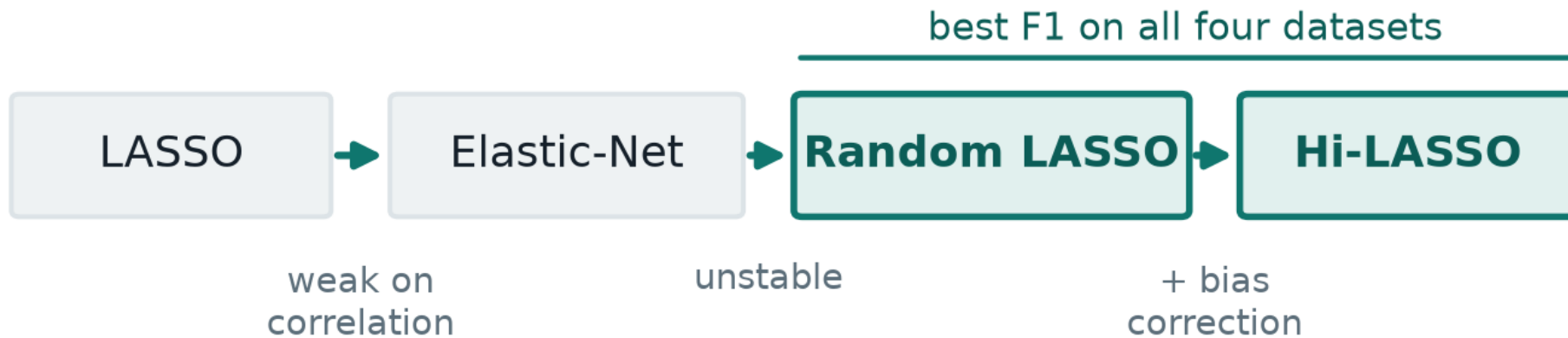
Real data has no answer key

No true β , so F1 can't be computed on real data.

→ Judged genes by coefficient & correlation.

Bootstrap beats plain LASSO — even in high dimensions

Bootstrap methods win — and stay strong even in high dimensions



Random LASSO and Hi-LASSO are most reliable, even at 10,000 genes. **Next: Stochastic LASSO and the real-data test.**

Sources and who did what

References

Methods

1. Tibshirani (1996) — LASSO. JRSS-B.
2. Wang et al. (2011) — Random LASSO. Ann. Appl. Stat.
3. Kim et al. (2019) — Hi-LASSO. IEEE Access.
4. Baek et al. (2026) — Stochastic LASSO. Sci. Rep.

Gene validation

5. ONECUT3 — Zupančič et al. (2024).
- 6–8. HILS1 — Lee (2022); Wang (2019); Liu (2019).
- 9–11. COL22A1 — Yan (2025); Liu (2023); Barbosa (2024).

Contribution

- **Boseung Jung**
Investigation (experimental settings), Analysis, Visualization (presentation materials), Literature Survey
- **Seongyun Ryu**
Conceptualization (Background & Significance), Investigation (experimental settings), Methodology, Analysis, Report Writing
- **Minseo Lee**
Conceptualization (Background & Significance), Analysis, Visualization (presentation materials), Literature Survey
- **Axel Botticelli**
Conceptualization (Background & Significance), Investigation (experimental settings), Analysis, Visualization (results)